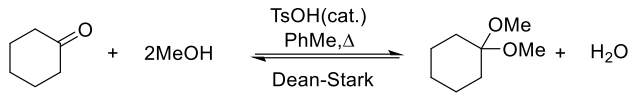
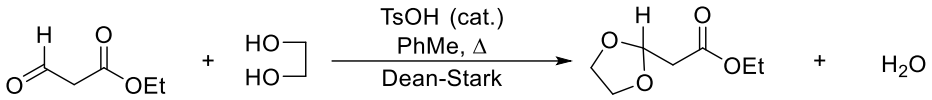
Class 9

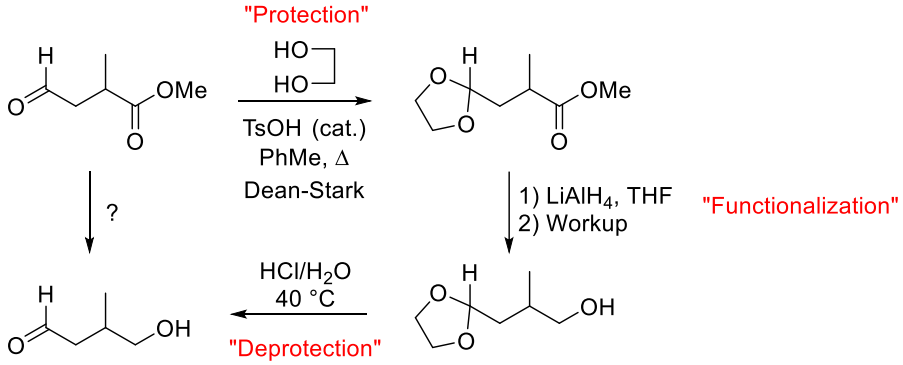
General procedure of acetal formation:



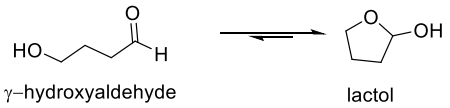
- Entropically less favorable.
- Use diols to make the reaction entropy neutral.

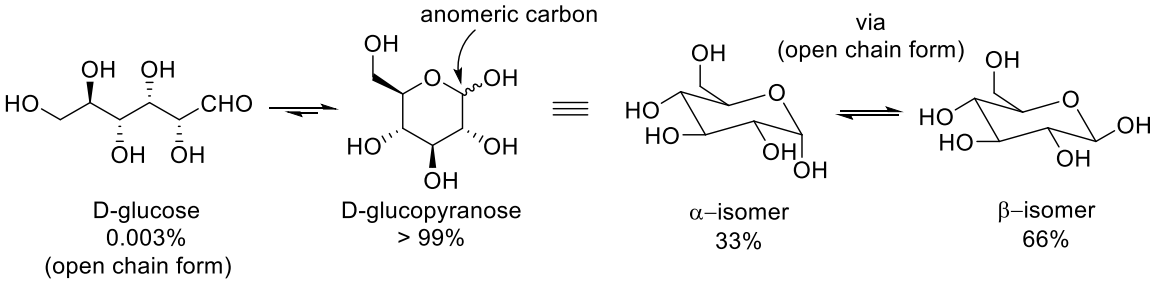


- Aacetals are stable in neural and basic conditions, can easily be hydrolyzed in aqueous acidic conditions. The mechanism is exactly the opposite.
- The five-membered cyclic acetal is known as a dioxolane. They are more stable than the acyclic ones. (Why?)
- Acetals can be used as the **protecting group**.



Hemi-acetals are less probable than the acetals. However, intramolecular processes make them favorable.

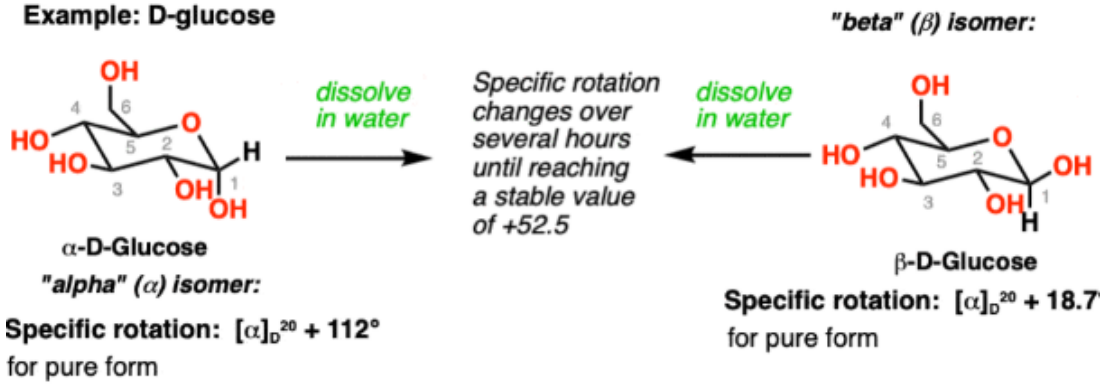




Mutarotation

- When pure alpha- and beta- anomers of sugars are dissolved in a solvent, their measured specific rotations will change over time, until converging on an intermediate value.
- This change in observed specific rotation is called, "**mutarotation**".

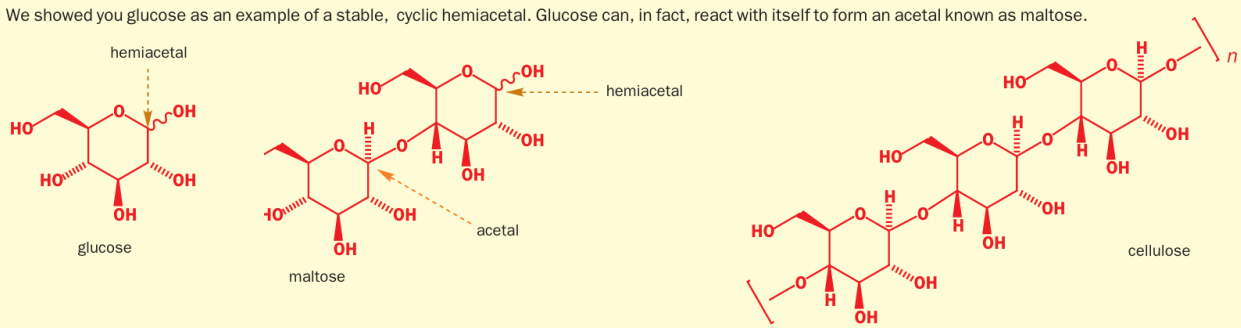
Example: D-glucose



- This intermediate value reflects an **equilibrium mixture** of the alpha- and beta-anomers, which can interconvert through the intermediate **open-chain** form.

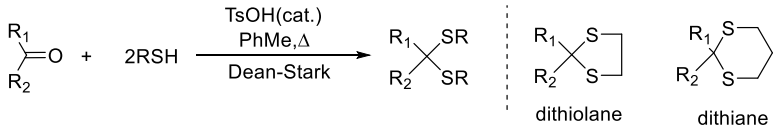
Acetals in nature

We showed you glucose as an example of a stable, cyclic hemiacetal. Glucose can, in fact, react with itself to form an acetal known as maltose.



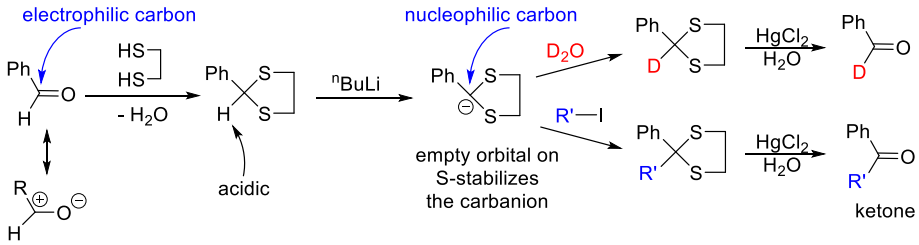
Maltose is a disaccharide (made of two sugar units) produced by the enzymatic hydrolysis of starch or cellulose, which are themselves polyacetals made up of a string of glucose units.

Thio acetals/ ketals:

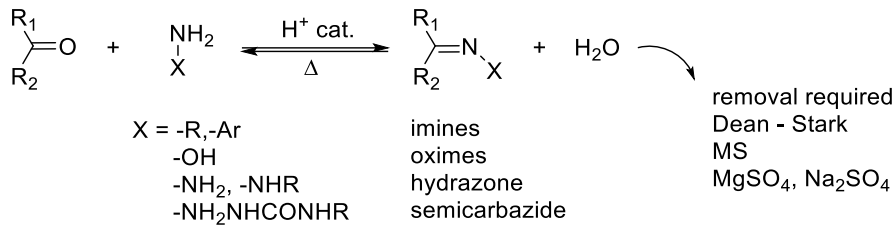


- S – center better nucleophile thus formation is faster
- S – center is soft so deprotection required soft Lewis acid (thiophilic), e.g. $HgCl_2$, $CdCO_3$ etc.

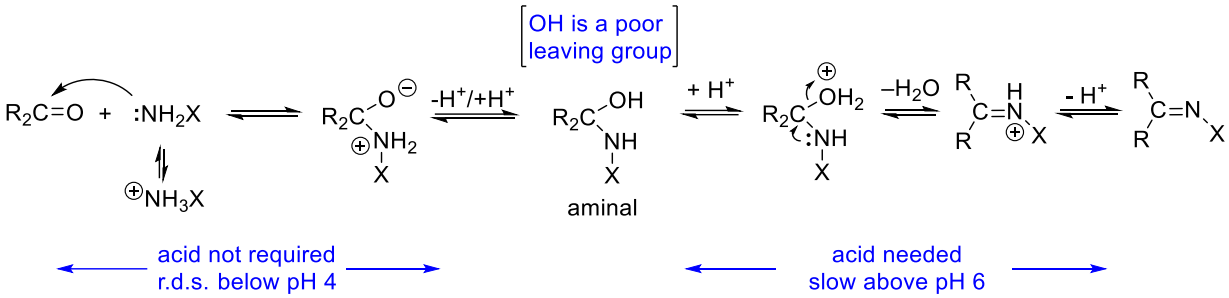
Application: (Reversal of polarity = Umpolung)



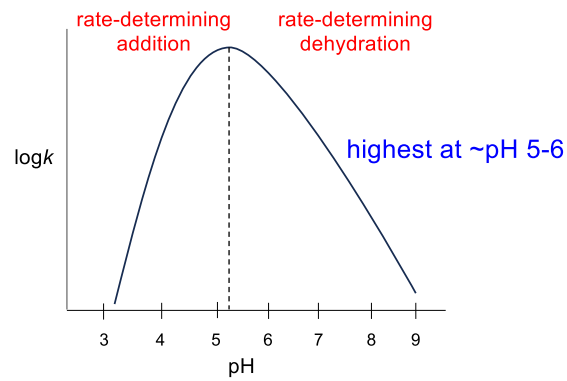
Reactions with nitrogen nucleophiles

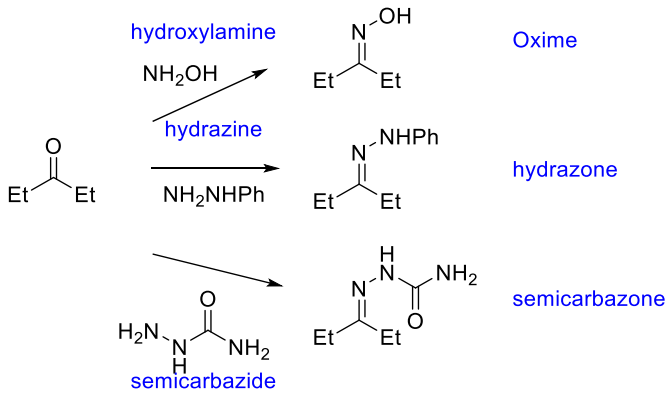


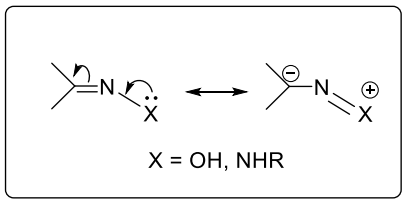
General mechanism:



- Catalyzed by acid.
- Rate increases with increasing acidity
- Highest at pH~5-6
- Below pH:5 rates slow as most amines are protonated.
- Above PH:7 rates slow as the dehydration step is slow
- Imines are sensitive to acid/H_2O (hydrolyzed). (mechanism is exactly opposite)
- Oximes and hydrazones are less prone to hydrolysis.



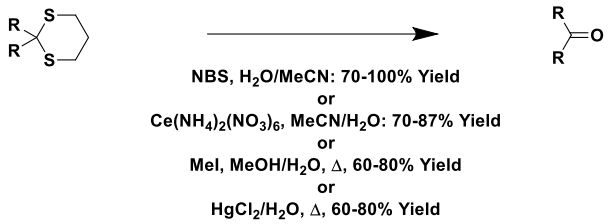




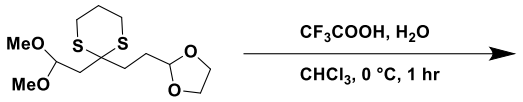
Delocalization decreases the δ+ charge on the carbon atom of the imine double bond and raises the energy of the LUMO, making it less susceptible to nucleophilic attack.

Questions:

1. A thioketal can be cleaved by the following reagents. Provide a plausible mechanism.

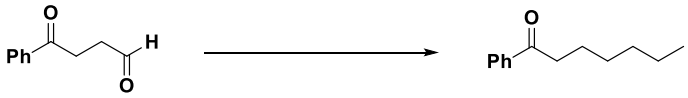


2. Predict the product (here acetals are present as protecting group)

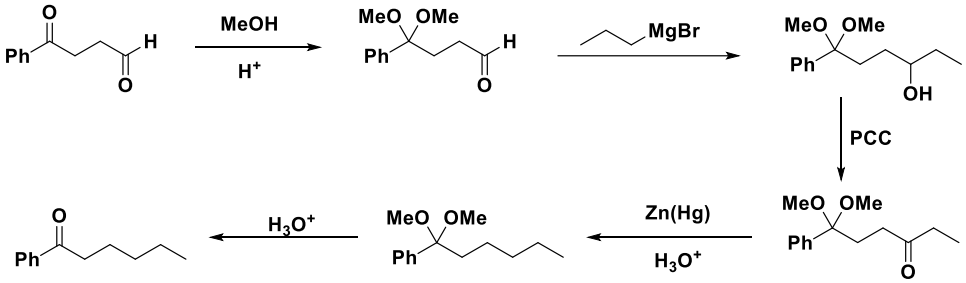


3. Write the product for the following reaction with mechanism:

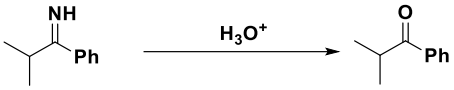
4. For the following synthesis identify the error(s) in the sequence shown that prevent them from being successful. Rework the synthesis so you have the right order/reagents.



Find the flaws:

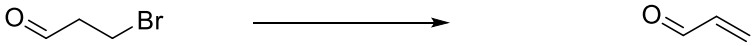
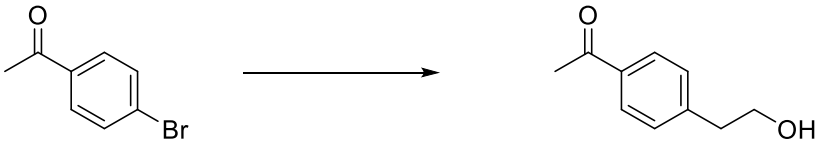


5. Write the product of the following mechanism, ii) Predict pH vs Rate plot (b/w 0-10 pH scale)



6. Please carry out the following transformation.

(i)



(i)